Ayudantía 2

Problema 1 - n° adimensionales

Parámetros del problema: F, ℓ, V, ρ, μ, c (6)

↳ Dimensiones: $F: [MLT^{-2}]$ $\quad V: [LT^{-1}]$ $\quad \mu: [ML^{-1}T^{-1}]$

$\ell: [L]$ $\quad \rho: [ML^{-3}]$ $\quad c: [LT^{-1}]$

· 3 dimensiones: (M, L, T)

Parámetros - dimensiones = $n\pi$: $6 - 3 = 3$ n° adimensionales

Teorema Pi

↳ Bases: V, ρ, ℓ (Deben ser l.i. y no contener la variable estudiada ~ F)

1. $[M^0 L^0 T^0] = [MLT^{-2}][LT^{-1}]^{\alpha}[ML^{-3}]^{\beta}[L]^{\gamma}$ $\quad \sim F$

$$\left.\begin{array}{l} M: 1+\beta = 0 \\ T: -2+\alpha = 0 \\ L: 1+\alpha-3\beta+\gamma = 0 \end{array}\right\} \begin{array}{l} \alpha = -2 \\ \beta = -1 \\ \gamma = -2 \end{array} \qquad \pi_1 = \frac{F}{(V\ell)^2 \rho}$$

2. $[M^0 L^0 T^0] = [ML^{-1}T^{-1}][LT^{-1}]^{\alpha}[ML^{-3}]^{\beta}[L]^{\gamma}$ $\quad \sim \mu$

$$\left.\begin{array}{l} M: 1+\beta = 0 \\ T: -1-\alpha = 0 \\ L: -1+\alpha-3\beta+\beta = 0 \end{array}\right\} \begin{array}{l} \alpha = -1 \\ \beta = -1 \\ \gamma = -1 \end{array} \qquad \pi_2 = \frac{\mu}{V\rho\ell}$$

3. $[M^0 L^0 T^0] = [LT^{-1}][LT^{-1}]^{\alpha}[ML^{-3}]^{\beta}[L]^{\gamma}$ $\sim c$

$$\begin{cases} M: \ \beta = 0 \\ T: \ -1 - \alpha = 0 \\ L: \ 1 + \alpha - 3\beta + \gamma = 0 \end{cases} \quad \begin{cases} \alpha = -1 \\ \beta = 0 \\ \gamma = 0 \end{cases}$$

$\pi_3 = \frac{c}{V}$

Significado físico:

1. $\pi_1 = \frac{F}{(Vl)^2 \rho}$: Fuerza de arrastre vs. fuerzas inerciales

2. $\pi_2 = \frac{\mu}{V \rho l} = \frac{1}{Re}$: n° de Reynolds

3. $\pi_3 = \frac{c}{V} = \frac{1}{Ma}$: n° de Mach

Problema 2 - nº adimensionales

Parámetros del problema: P, N, D, ρ, μ (5)

↳ Dimensiones: P: $N: [T^{-1}]$ $D: [L]$ $\rho: [ML^{-3}]$ $\mu: [ML^{-1}T^{-1}]$ (3)

$\pi: 5 - 3 = 2$ nº adimensionales

Bases: N, D, ρ

1. $[M^0 L^0 T^0] = [ML^2 T^{-3}][T^{-1}]^{\alpha}[L]^{\beta}[ML^{-3}]^{\gamma}$ ~ P

$M: 1 + \gamma = 0$
$T: -3 - \alpha = 0$
$L: 2 + \beta - 3\gamma = 0$

$\alpha = -3$
$\beta = -5$
$\gamma = -1$

$\pi_1 = \dfrac{P}{\rho N^3 D^5}$

2. $[M^0 L^0 T^0] = [ML^{-1}T^{-1}][T^{-1}]^{\alpha}[L]^{\beta}[ML^{-3}]^{\gamma}$ ~ μ

$M: 1 + \gamma = 0$
$T: -1 - \alpha = 0$
$L: -1 + \beta - 3\gamma = 0$

$\alpha = -1$
$\beta = -2$
$\gamma = -1$

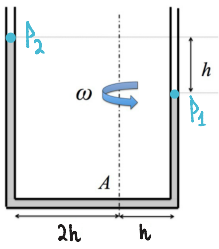
$\pi_2 = \dfrac{\mu}{\rho N D^2} = \dfrac{1}{Re}$

Lo relacionamos: $\pi_1 = \phi(\pi_2)$

$$\frac{P}{\rho N^3 D^5} = \phi\left(\frac{\mu}{\rho N D^2}\right)$$

$$P = \rho N^3 D^5 \cdot \phi\left(\frac{\mu}{\rho N D^2}\right)$$

Problema 2 - Eq. mecánico



Coordenadas cilíndricas: $\nabla = \left(\frac{\partial}{\partial r}, \frac{1}{r}\frac{\partial}{\partial \theta}, \frac{\partial}{\partial z}\right)$

Equilibrio mecánico: $\nabla p = \rho \cdot \vec{f_b}$

$\vec{f_b} = (r\omega^2, 0, -g)$

$\left(\frac{\partial p}{\partial r}, \frac{1}{r}\frac{\partial p}{\partial \theta}, \frac{\partial p}{\partial z}\right) = \rho(r\omega^2, 0, -g)$ → 3 ecuaciones:

1. $\frac{\partial p}{\partial r} = \rho r \omega^2$
2. $\frac{1}{r}\frac{\partial p}{\partial \theta} = 0$
3. $\frac{\partial p}{\partial z} = -\rho g$

Partimos con esta por conveniencia ↓

2. $\frac{1}{r}\frac{\partial p}{\partial \theta} = 0 \quad / \cdot r$, con $r \neq 0$

$\frac{\partial p}{\partial \theta} = 0 \quad / \int d\theta$

$\int dp = \int 0 \cdot d\theta$

$p(r, \theta, z) = \varphi(r, z)$ · Ahora sabemos que la presión varía según una función que depende de las variables (r, z) - y no del ángulo θ.

1. $\frac{\partial p}{\partial r} = \frac{\partial \varphi}{\partial r}$

$\frac{\partial \varphi}{\partial r} = \rho r \omega^2 \quad / \int dr$

$\int d\varphi = \int \rho r \omega^2 \, dr$

$\varphi(r, z) = \frac{1}{2}\rho r^2 \omega^2 + \psi(z)$ → Falta revisar el comportamiento en z.

3. $\frac{\partial p}{\partial z} = \frac{\partial \varphi}{\partial z} = \frac{\partial \psi}{\partial z}$ $\qquad \frac{\partial \psi}{\partial z} = -\rho g \quad / \int dz$

$$\int d\psi = -\int \rho g \, dz$$

$$\psi(z) = -\rho g z + C$$

Entonces:

$$p(r,\theta,z) = \varphi(r,z) = \frac{1}{2}\rho r^2\omega^2 + \psi(z) = \frac{1}{2}\rho r^2\omega^2 - \rho g z + C$$

- Punto 1 : (h, z_1)
- Punto 2 : $(2h, z_2)$

$\Big\}$ $z_2 - z_1 = h$

Igualamos ambos puntos, ya que tienen la misma presión (atmosférica).

$$\frac{1}{2}\rho h^2\omega^2 - \rho g z_1 + C = \frac{1}{2}\rho (2h)^2\omega^2 - \rho g z_2 + C$$

$$\rho h^2\omega^2 - \rho g z_1 = \rho (2h)^2\omega^2 - \rho g z_2$$

$$\rho g z_2 - \rho g z_1 = (2h)^2\omega^2 - \rho h^2\omega^2$$

$$\rho g (z_2 - z_1) = 3\rho h^2\omega^2$$

$$h = \frac{3h^2\omega^2}{g} \rightarrow \omega^2 = \frac{g}{3h}, \quad \omega = \sqrt{\frac{g}{3h}}$$